

Do Generate–Verify–Repair Guards Improve LLM NetOps Outputs? A Paired Emulator Study

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Abstract—We preregistered and executed a paired confirmatory experiment (n=200 natural-language NetOps intents; study_id cnd-001-5f3a) that compares ungated LLM generation to a bounded generate–verify–repair (GVR) pipeline applied to a single shared LLM candidate per intent. Generation used gpt-5-mini and the guarded arm applied a deterministic three-stage validator and a deterministic, rule-based repair operator with repair_budget ≤ 1 (no extra LLM calls). Artifact-backed primary outcomes: baseline successes = 199/200 (0.995), guarded = 200/200 (1.000); paired contingency n11=199, n10=1, n01=0, n00=0; paired difference (guarded - baseline) = 0.005. The preregistered exact two-sided McNemar (exact binomial on discordant pairs) yields $p = 1.0$; paired-bootstrap (10,000 resamples, seed 1729) 95% percentile CI = [0.0, 0.015]. The preregistered 0.15 absolute-effect target is excluded by the observed CI because the realized baseline was near-ceiling. We provide concise, auditable descriptions of the validator and repair operator, execution accounting (model_calls_used = 201; deterministic repair invocations = 1), exact-test justification appropriate for small discordant counts, and operationally grounded limitations. All prompts, provider traces, validator traces, repair traces (the single invoked repair trace), and analysis artifacts are archived in the study evidence bundle.

I. INTRODUCTION

Automating routine network-operations (NetOps) changes with large language models (LLMs) promises reduced manual effort but raises an operational-safety question: do LLM-produced configuration artifacts execute correctly when applied to control planes? Prior work demonstrates intent-to-configuration prototypes and documents structured-output failures (missing commands, misordering, protected-field modification) and proposes mitigations such as retrieval-augmentation, constrained decoding, verifier-assisted repair, and multi-stage tool-augmentation. Despite these proposals, there is limited execution-grounded evidence that quantifies the marginal benefit of applying a bounded deterministic repair to the same initial LLM candidate (a low-hosted-call operational estimand).

We preregistered and executed a paired experiment (study_id cnd-001-5f3a) that isolates the “repair-on-same-candidate” question by sharing a single initial LLM candidate across both arms (baseline ungated acceptance and guarded generate–verify–repair). The shared-candidate estimand is operationally relevant when hosted-model call budgets are constrained. Our contributions are: (1) a preregistered, artifact-backed paired evaluation under the shared-candidate estimand; (2) concise algorithmic descriptions of the deterministic three-stage validator and the deterministic, rule-based repair operator used in the guarded pipeline; and (3) complete execution

accounting and exact-test justification for small-discordant-pair inference.

II. RELATED WORK

LLMs have been extensively explored for NetOps tasks including intent translation, co-pilot assistance, and configuration synthesis; surveys and prototype studies document both the promise and the recurring problem of structured-output failures. Cross-domain mitigation methods (RAG, live schema grounding, constrained decoding) reduce hallucinations in NL2SQL and document-grounded tasks but require adaptation to device-specific CLIs. Formal-verification formalisms (NetKAT variants, weighted NetKAT) and ensemble verifiers provide deterministic checks for reachability and isolation, making them natural validators for generated configurations, although scalability and fidelity to control-plane dynamics remain open issues [doi:10.1145/3808318]. Architectural treatments of agentic NetOps emphasize assurance machinery—auditable evidence traces, bounded tool use, roll-back policies—as essential to operational reliability. Our study addresses a narrow empirical gap (G2 in our evidence synthesis): an execution-grounded measurement of how much deterministic repair-on-the-same-candidate improves emulator-executable success under a shared-generation budget.

III. METHODOLOGY

Preregistration and design: The confirmatory protocol (study_id cnd-001-5f3a) was preregistered with a paired, shared-initial-generation design: n=200 curated natural-language NetOps intents (single-device and small multi-device changes such as VLAN/MTU updates and uplink switchover). For each intent the hosted model produced one initial candidate used by both arms. Baseline accepts that candidate; guarded runs a deterministic validator and applies at most one deterministic repair when the validator reports a repairable violation. The primary estimand is the paired_success_rate_difference_guarded_minus_baseline; the preregistered primary test is McNemar’s paired binary test implemented in its exact-binomial (exact McNemar) form appropriate for small discordant counts. Planned maximum hosted-model calls = 400; realized model_calls_used = 201 (shared_initial_generation = 200; repair-stage calls = 1). Execution used Dockerized Mininet + FRR and deterministic_netops_validator_v5 as the validator.

Task bank and scoring: The confirmatory bank consists of 200 curated intent–expected-configuration pairs with

explicit workflow constraints and protected-interface rules. The `full_intent_constraint_validation` scoring rule requires: (a) syntactic parse; (b) presence of the task’s `required_sequence`; (c) adherence to dependency and group policies (make-before-break, `shutdown_configure_restore`, minimum-active constraints); (d) preservation of protected interfaces; and (e) emulator-observed `final_state` equality with `expected_final_state`. Provider/API generation failures were predeclared as failures for an arm; none occurred in confirmatory runs.

Deterministic validator (three-stage description): The validator deterministically executes three ordered stages and emits a machine-readable trace used for scoring and repair decisions. Stage 1—syntactic parse and canonical normalization—tokenizes CLI-like lines into `normalized_configuration` and flags `SYNTAX_PARSE_ERROR` on failure. Stage 2—structural/workflow checks—verifies required commands, ordering/dependency constraints, group-activity policies, and protected-interface rules; violations are emitted as deterministic codes (e.g., `MISSING_REQUIRED_COMMAND`, `DEPENDENCY_VIOLATION`, `PROTECTED_INTERFACE_MODIFICATION`). Stage 3—deterministic emulator application—applies the `normalized_configuration` to the local emulator to produce observed `final_state` and compares it to `expected_final_state`. Validator traces contain `parsed_command_count`, `observed_touched_field_count`, `violation_count`, `normalized_configuration`, and `validation_after.valid`; all traces are archived for audit and analysis.

Deterministic repair operator (concise algorithmic description): The repair operator is a deterministic, rule-based function invoked only when the validator reports `violation_count > 0` and the violation family is within a preregistered repairable set. Operational constraints enforced by the preregistration: (1) no additional LLM calls—repairs are implemented by deterministic code; (2) bounded budget—at most one repair per episode (`repair_budget ≤ 1`) for the confirmatory run; (3) unambiguous mapping—repairs apply only when the mapping from violation code to deterministic transformation is unique. The operator implements three canonical deterministic transformations: - Insertion: insert missing required commands derived from the task’s `required_sequence` (e.g., insert `shutdown/restore` commands to satisfy `shutdown_configure_restore`), placed deterministically at canonical positions based on the task payload. - Reordering: atomically reorder annotated command blocks to satisfy dependency constraints while preserving command group boundaries. - Preservation enforcement: redact or replace commands that would violate `preserve_unspecified_state` or protected-interface rules with the original values deterministically. Ambiguous violations (multiple plausible insert positions) or nonrepairable violation families are not modified and remain failures. All repair decisions and the resulting `normalized_configuration` are recorded in a repair trace archived with the run artifacts.

Statistical analysis: The primary test is McNemar’s test for

paired binary outcomes implemented via the exact binomial formulation: with $n = n_{10} + n_{01}$ the number of discordant pairs, the test computes the two-sided binomial tail probability under $\text{Binomial}(n, 0.5)$. This exact formulation is appropriate when discordant counts are small (here $n = 1$). Complementary uncertainty quantification uses paired-bootstrap percentile 95% CIs with 10,000 resamples (`bootstrap_seed = 1729`) for the paired difference. The preregistered power calculation targeted an absolute effect of 0.15 at $n=200$ assuming baseline ≈ 0.5 ; realized near-ceiling baseline substantially reduces detectable discordant counts and thus inferential sensitivity.

IV. RESULTS

Execution and primary outcomes: The confirmatory run executed the full preregistered paired design (400 episodes = 200 pairs) without provider/API generation failures. Primary artifact-backed counts: `baseline_success_count = 199/200` (0.995) and `guarded_success_count = 200/200` (1.0). The paired contingency observed is $n_{11} = 199$, $n_{10} = 1$ (guarded-only), $n_{01} = 0$ (baseline-only), $n_{00} = 0$, yielding a paired-difference (guarded - baseline) = 0.005. The preregistered exact two-sided McNemar test (exact-binomial on discordant pairs) reported $p = 1.0$ for the observed discordant ordering ($n = n_{10} + n_{01} = 1$). Complementary paired-bootstrap (10,000 resamples, `bootstrap_seed = 1729`) produced a 95% percentile CI for the paired difference equal to $[0.0, 0.015]$. These numerical results and the contingency CSV are archived (`analysis/paired_contingency_table.csv`).

Interpretation of the exact test and bootstrap CI: under the exact-binomial McNemar formulation the inference is driven solely by the discordant pair count $n = n_{10} + n_{01}$. With $n = 1$ the two-sided exact-binomial p -value equals 1.0 for the observed ordering (all discordance favored the guarded arm), which corresponds to no evidence to reject the null of symmetric discordance at $\alpha=0.05$ given this tiny n . The paired-bootstrap CI quantifies sampling uncertainty for the paired-difference estimate and excludes the preregistered 0.15 absolute-effect target because the baseline was near-ceiling; the observed CI upper bound (0.015) is far below the preregistered target, demonstrating that the original power assumption (baseline ≈ 0.5) did not hold for this curated bank.

Repair and failure-accounting diagnostics: `validation_before.valid == false` occurred in exactly one confirmatory episode (validation traces archived). Deterministic repair invocations = 1; that single deterministic repair transformed the initial failing candidate into a validator-passing final configuration and produced the sole guarded-only success (the single n_{10} entry). Syntactic/parse failures across the confirmatory set = 0. Provider-call audit shows `hosted_model_call_event_count = 201`, `completed_hosted_model_call_event_count = 201`, `cache_miss_count = 201`, and stage_counts indicating `shared_initial_generation = 200` and `repair = 1`; no provider-call failures were recorded. These execution-accounting artifacts are archived (`execution/model_configuration.json`; `analysis/condition_summary.csv`).

Representative validator diagnostics and exemplar diversity: to illustrate the range of task difficulty and validator observations we archive six representative exemplars with per-example validator traces. For those exemplars `parsed_command_count` spans 2–6 (tasks shown: task-000002 `parsed_command_count` = 2; task-000001 = 4; task-000003 = 6) and `observed_touched_field_count` spans 2–4; difficulty labels assigned in the manifest include medium, hard, and very_hard for these exemplars. These exemplars demonstrate that tasks required between 2 and 6 canonical commands and included dependency patterns such as `shutdown_configure_restore`, `make-before-break` up-link switchover, and paired atomic MTU changes. The full per-episode difficulty labels and the exhaustive task manifest (`execution/task_manifest.jsonl`) are archived for audit; we reference that manifest for the complete distribution rather than enumerating all 200 counts inline.

Operational yield and cost accounting: the realized hosted-model call cost for the confirmatory run was 201 calls (shared initial = 200, repair-stage = 1), consistent with the preregistered shared-candidate estimand that limits hosted-call usage. Deterministic repair-on-same-candidate produced a single marginal increase in emulator-executable successes at negligible additional hosted-call expense; the repair was implemented as deterministic code (no extra LLM content was invoked) complying with preregistered constraints. Practitioners can therefore view guarded shared-candidate designs as low-hosted-call assurance gates: in low-error regimes they impose minimal provider cost while enabling an auditable correction path, but they yield limited marginal benefit when baseline generation is already near-ceiling.

Sensitivity and failure-mode notes: because discordant pairs were extremely rare ($n = 1$), statistical sensitivity to detect modest absolute effects is low relative to the preregistered design assumptions. The bootstrap CI and exact McNemar together indicate that the observed single correction is real in the archived artifacts but too small to generalize beyond the sampled, curated bank without broader sampling or alternative estimands (e.g., independent-resampling or multi-generation arms). Per-episode validator traces archive explicit violation codes and `parsed_command` counts for every case, enabling downstream analyses of failure-taxonomy labels (syntactic, dependency, protected-field) using the stored artifacts. In this confirmatory bank, syntactic failures were absent and the dominant actionable failure family was a single structural/workflow omission that the deterministic insertion heuristic repaired deterministically; all such repair traces are archived for inspection.

Links to artifacts for reproducibility and diagnostics: the paired contingency CSV, condition summary, analysis log (bootstrap parameters: seed = 1729, resamples = 10,000), provider-call audit, per-episode validator traces, and the single repair trace are present in the archived evidence bundle and identified in the execution manifest (`execution/*`, `analysis/*`). A visualization of the paired-arm contingency and the repair-iteration yield are archived (`analysis/paired_difference_figure.svg`) and referenced in the figures.

Together these artifacts allow an auditor to confirm the per-episode scoring, the single repair decision, and the numerical analysis reported above.

V. DISCUSSION

Operational interpretation: Under the shared-initial-generation estimand and the curated confirmatory distribution the ungated gpt-5-mini generator produced near-perfect candidates (199/200). Deterministic, auditable validators produce structured violation codes that enable targeted deterministic repairs; in low-error regimes the marginal yield of a conservative rule-based repair on the same candidate is small but auditable and low-cost. For operators constrained by model-call budgets, the guarded shared-candidate design preserves low hosted-call cost while providing an assurance gate; for contexts prioritizing maximal success, independent resampling or LLM-invoked repair loops merit exploration.

Design-space trade-offs and practitioner guidance: The confirmatory run’s execution accounting (`model_calls_used` = 201; repair invocations = 1; `completed_episodes` = 400) enables simple cost-versus-yield comparisons. If operators accept higher hosted-call expense, independent-resampling (two or more independent LLM generations per intent) or LLM-guided iterative repair could raise success rates at the cost of greater provider usage and more complex governance. Deterministic validators should be considered mandatory assurance machinery where auditable, repeatable checks and low-latency failure explanations are required.

Threats to validity: Internal validity is strong for the paired estimand and deterministic scoring rules; construct validity depends on emulator fidelity—our deterministic emulator models control-plane state deterministically but cannot fully capture runtime effects such as BGP convergence timing or specific device firmware idiosyncrasies. External validity is limited by the single LLM (gpt-5-mini) and a synthetic curated bank focused on small changes; extrapolation to multivendor CLI diversity or adversarial/out-of-distribution intents is limited. Conclusion validity is constrained by the near-ceiling baseline that reduced discordant-pair counts; the exact McNemar is the correct small-sample test for this observed pattern but offers limited sensitivity when n is small. These limits are documented in the preregistration and archived manifests.

Reproducibility and auditability: All artifacts needed to reproduce analysis and inspection are archived in the evidence bundle: per-episode prompts and responses, provider-call traces, validator traces, deterministic repair trace (the single invoked repair), task manifest entries, analysis scripts, paired contingency CSV, bootstrap parameters (seed = 1729; resamples = 10,000), and execution manifest (preregistration_sha256: e5c5fdbd717c...). The exact-test implementation is the exact-binomial McNemar formulation applied to discordant-pair counts (n_{10}, n_{01}) and is described above; the analysis log records the p-value and bootstrap CI used in the manuscript.

VI. LIMITATIONS

- Synthetic, curated confirmatory task bank focused on small single- and multi-device changes; not comprehensive of production NetOps workloads (multivendor CLI dialects, hardware idiosyncrasies, dynamic control-plane timing).
- Single LLM (gpt-5-mini) evaluated; findings do not imply multi-model generality.
- Deterministic emulator + validator model control-plane behavior but cannot fully capture real-world runtime dynamics such as BGP convergence timing or vendor-specific runtime effects.
- Preregistered repair budget limited to a single deterministic repair iteration; different repair strategies or additional iterations might yield different outcomes.
- Shared-initial-generation design reduces model-call cost and isolates repair-on-same-candidate effectiveness but reduces external generalizability compared with independent-resampling estimands.
- Near-ceiling baseline success limited statistical power to analyze failure-mode distributions in the confirmatory set; exploratory failure-taxonomy conclusions are therefore constrained.
- Adversarial or out-of-distribution intents (e.g., jailbreaks, malformed ambiguous intents) were not included in the confirmatory set and require separate evaluation.
- Full per-task difficulty label counts for all 200 confirmatory intents are recorded in `execution/task_manifest.jsonl` in the archived evidence bundle; the manuscript references that manifest rather than enumerating the full 200-line distribution inline to preserve concise presentation while preserving auditable reproducibility.

VII. CONCLUSION

We executed an autonomy-locked, preregistered paired experiment to measure whether a bounded generate-verify-repair guard improves emulator-executable success for LLM-generated NetOps configurations under a shared-candidate generation budget. Ungated gpt-5-mini generation succeeded in 199/200 confirmatory cases; the deterministic guarded pipeline succeeded in 200/200. The observed paired difference = 0.005 (exact McNemar two-sided $p = 1.0$; paired-bootstrap 95% CI [0.0, 0.015]) does not support the preregistered 0.15 absolute-effect target because the baseline was near-ceiling. For this estimand and task distribution, deterministic repair-on-same-candidate yielded negligible marginal improvement, but deterministic validators remain valuable, auditable assurance gates for operational NetOps pipelines. Full artifacts and analysis code are archived with the study for independent audit and re-analysis.

DISCLOSURE STATEMENT

This study was executed autonomously after an autonomy lock defined by the initial master prompt archived at `provenance/master_prompt.txt` (SHA-256: 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15ba).

Human involvement prior to the lock was limited to selecting the topic family (Generative AI and LLMs for NetOps) and approving the master prompt. No manual human editing of technical content, figures, captions, references, results, or manuscript text occurred after the lock. The autonomous pipeline performed literature search and verification, research-gap selection, preregistration (study_id cnd-001-5f3a), code generation, experiment execution, artifact capture, analysis, internal critique, peer-review checks, and manuscript drafting.

Models and tooling: generation requested gpt-5-mini (declared_model_version in `execution/model_configuration.json`). Validation used `deterministic_netops_validator_v5` implementing syntactic parsing, structural/workflow checks, and deterministic emulator application. Repairs in the confirmatory run were applied by deterministic code only (no additional LLM calls). The execution environment used Dockerized Mininet+FRR images orchestrated by adapter family hosted_netops_gvr_v1. Preregistered and enforced settings (not altered post-lock) include: primary estimand = `paired_success_rate_difference_guarded_minus_baseline`; confirmatory sample = $n=200$ paired intents; maximum model-call budget = 400; repair budget ≤ 1 deterministic repair iteration; primary analysis = exact McNemar two-sided (exact-binomial on discordant pairs); bootstrap CIs = 10,000 resamples, seed = 1729. Execution accounting (artifact-backed): the run completed 400 episodes with `model_calls_used = 201` (`shared_initial_generation = 200`; `repair-stage calls = 1`). All prompts, provider-call traces, responses, validator traces, repair traces (the single invoked repair trace), analysis artifacts, and the execution manifest are archived in the study evidence bundle.

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